

Publication 1

Field	Description
Title	Churn Buster: Uncovering Patterns and Predicting Churn in OTT Platforms
Harvard Reference	Authors unknown. (2024) <i>Churn Buster: Uncovering Patterns and Predicting Churn in OTT Platforms</i> . IEEE Conference Publication. Available at: https://ieeexplore.ieee.org/abstract/document/10716193 (Accessed: 20 September 2026). [ieeexplore.ieee.org]
Link	https://ieeexplore.ieee.org/abstract/document/10716193
Modelling	Machine learning-based churn prediction models were used to identify customers likely to leave OTT platforms. The study focused on predictive analytics to classify customers according to churn risk and support retention decision-making. [ieeexplore.ieee.org]
Preprocessing Techniques	Customer behaviour data were prepared through data cleaning, feature extraction, customer segmentation, and engagement analysis. Behavioural variables were transformed into features suitable for predictive modelling.
Datasets Used	OTT platform subscriber data containing customer engagement, usage behaviour, subscription activity, and churn-related information. [ieeexplore.ieee.org]
Key Takeaways	1. Customer engagement behaviour is a strong predictor of churn. 2. Early identification of at-risk subscribers enables proactive retention strategies. 3. Behavioural data should be included when developing churn prediction models.

This study is highly relevant because STADIOchoice is an OTT streaming provider facing increasing subscriber churn. The focus on subscriber behaviour and retention closely aligns with the objectives of the STADIOstream churn prediction project.

Publication 2

Field	Description
Title	A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI
Harvard Reference	Authors unknown. (2024) <i>A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI</i> . IEEE Journal Publication. Available at: https://ieeexplore.ieee.org/abstract/document/10530632 (Accessed: 20 September 2026). [ieeexplore.ieee.org]
Link	https://ieeexplore.ieee.org/abstract/document/10530632
Modelling	A hybrid machine learning approach was used to predict customer churn. The study incorporated Explainable Artificial Intelligence (XAI) techniques to improve model transparency and help identify the drivers of churn. [ieeexplore.ieee.org]
Preprocessing Techniques	Big data preparation techniques, feature engineering, data transformation, and customer behaviour analysis were applied before model development.
Datasets Used	Streaming service customer data containing subscriber activity, behavioural metrics, customer engagement information, and retention-related variables. [ieeexplore.ieee.org]
Key Takeaways	1. Explainable AI helps organisations understand why customers churn. 2. Hybrid models can improve churn prediction performance. 3. Data-driven retention strategies can improve customer retention outcomes.

This publication aligns strongly with STADIOchoice because the organisation wants to understand why subscribers stay or leave. Explainable AI could help management understand which factors contribute most to subscriber churn and support evidence-based retention strategies.

Publication 3

Field	Description
Title	Customer Churn and Content Strategy Analysis for Netflix
Harvard Reference	Authors unknown. (2023) <i>Customer Churn and Content Strategy Analysis for Netflix</i> . In: Springer Nature Conference Proceedings. Available at: https://link.springer.com/chapter/10.1007/978-3-032-31998-2_33 (Accessed: 20 September 2026).
Link	https://link.springer.com/chapter/10.1007/978-3-032-31998-2_33
Modelling	Ensemble machine learning models including XGBoost, CatBoost, Random Forest, and AdaBoost were used to predict customer churn. SHAP (SHapley Additive Explanations) analysis was applied to explain model predictions and determine feature importance.
Preprocessing Techniques	Customer engagement analysis, watch behaviour analysis, subscription trend analysis, feature engineering, and SHAP-based feature importance analysis were conducted before modelling.
Datasets Used	Netflix customer churn and engagement data including viewing behaviour, subscription trends, watch time, customer engagement metrics, content performance data, and regional viewing preferences.
Key Takeaways	1. Ensemble models such as XGBoost and Random Forest are highly effective for churn prediction. 2. Average watch time per day and total watch hours are the strongest predictors of churn. 3. Customer retention can be improved by combining churn prediction with content strategy decisions.

This publication is particularly relevant because STADIOchoice operates a streaming platform similar to Netflix. The findings demonstrate how subscriber engagement and viewing behaviour can be used to predict churn and improve content investment decisions.

Publicly Available Dataset for Part B

Field	Description
Name	Netflix Customer Churn Dataset
Harvard Reference	Abdul Wadood (2024) <i>Netflix Customer Churn Dataset</i> . Kaggle. Available at: https://www.kaggle.com/datasets/abdulwadood11220/netflix-customer-churn-dataset (Accessed: 24 September 2026).
Link	https://www.kaggle.com/datasets/abdulwadood11220/netflix-customer-churn-dataset
Number of Instances	To be confirmed after downloading the dataset. Report the exact number of records contained in the dataset.
Number of Feature Columns	To be confirmed after downloading the dataset. Report the original number of feature columns available before preprocessing.
Number of Classes	2 classes: Churned and Not Churned.
Target Range (if applicable)	Not applicable because this is a classification problem.
Origin	The dataset is publicly available on Kaggle and was created for customer churn prediction in a Netflix-style streaming environment. It contains subscriber information, engagement metrics, and customer-related variables that can be used to identify factors associated with subscriber retention and churn.

Dataset Justification

The Netflix Customer Churn Dataset is an appropriate choice because it focuses on a streaming subscription service and contains customer engagement and churn information that closely aligns with STADIOchoice's business environment. The dataset supports the development of churn prediction models and enables the analysis of subscriber behaviour, making it highly relevant to the STADIOstream Subscriber Churn Prediction Project.